Zhankun Luo

Machine Learning Researcher & Engineer

Optimization | Statistical Learning | Computer Vision

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SUMMARY

Ph.D. researcher and ML engineer building optimization and computer-vision systems. Develops Python/C++ research software and ML pipelines from data preparation to plant deployment; published at ICML, UAI, TMLR, and CVPR Workshops.

EDUCATION

Purdue University - Ph.D. in Electrical and Computer Engineering **Present**
Purdue University Northwest - M.S. in Electrical and Computer Engineering **2021**
Beijing Institute of Technology - B.S. in Telecommunication Engineering **2019**

TECHNICAL EXPERTISE

Programming	Python, C/C++, SQL, MATLAB, Bash, C#, Java
ML / Scientific	PyTorch, TensorFlow, LightGBM, scikit-learn, NumPy, SciPy, pandas, Mathematica
Vision / Graphics	OpenCV, image registration, multi-view geometry, camera calibration, Qt, OpenGL, Unity
Engineering	Linux, Git, Docker, CMake, gdb; data pipelines, model evaluation, reproducible experimentation
Research	Optimization, statistical learning, federated learning, EM and latent-variable models, synthetic data
Coursework	Statistical ML; Probabilistic ML; Reinforcement Learning; Optimization for Deep Learning; Model-Based Image and Signal Processing; Estimation Theory; Digital Signal Processing; Pattern Recognition; Object-Oriented Design and Programming; Mathematical Statistics; Real Analysis; Linear Algebra

EXPERIENCE

Doctoral Researcher | MINDS Group, Purdue University **May 2023 – Present**
Optimization and statistical ML; Research Assistant since Aug. 2025 RCR certificate

- **Distributed optimization.** Designed and analyzed a single-loop switching-gradient method for constrained worst-client optimization; implemented full- and partial-client experiments within the NSF-funded Learning through the Air project. [1]
- **Statistical learning.** Built Python/Mathematica experiments and visualizations for EM dynamics and finite-sample behavior [2] [3] [11]; analyzed connections between score matching, maximum likelihood, and EM in mixed linear regression [10].
- **Variance reduction and randomized methods.** Established high-probability bounds for variance-reduced estimators [8]; studied randomized midpoint methods for constrained optimization [9] and optimal-transport antithetic sampling [12].

Research Assistant | VIPER Laboratory, Purdue University **Aug. 2021 – May 2022**
Computer vision, generative models, and fine-grained classification

- **Synthetic imagery for plant phenotyping.** Built an image-to-image GAN pipeline for PhenoSorg, generating 1,000 labeled, high-resolution samples from 400 real UAV images; evaluated augmentation for panicle detection and counting. [4]
- **Hierarchy-aware recognition.** Reimplemented PyTorch embeddings for the TADA dietary-assessment project on 15,000 images across 82 food categories; reduced average hierarchical distance by 3% at top-1 and 10% at top-5. code

Research Assistant | CIVS, Purdue University Northwest **Feb. 2020 – May 2021**
Industrial machine learning and computer vision

- **Smart Ladle.** Built DNN and LightGBM casting-temperature predictors using SQL process data and a Unity/C# interface; supported plant testing and deployment at Steel Dynamics' Butler Division. [5] project code
  Paper awards: AIST Digitalization Best Paper (2021); Hunt-Kelly Outstanding Paper, third place (2022).
- **Structured-light 3D vision.** Built a multi-camera, multi-laser C++/MATLAB/OpenCV measurement pipeline using U-Net segmentation and multi-level RANSAC for robust laser-stripe extraction and reconstruction. [6] [7] slides

ADDITIONAL MACHINE LEARNING PROJECTS

Fetal-Brain MRI Segmentation | Python, PyTorch, MATLAB
Implemented U-Net fetal-brain segmentation; evaluated FPN, adversarial-loss, and focal-loss variants. Annotated 6,700 MRI slices across three anatomical planes to support 3D registration. related abstract

Retinal-Vessel Segmentation Benchmark | Python, PyTorch, TensorFlow
Benchmarked SegCaps, U-Net, and DenseNet on DRIVE retinal-vessel segmentation; compared training convergence, overlap metrics, and predicted vessel masks. project dataset

SELECTED TECHNICAL PORTFOLIO

Smart Ladle | Python, LightGBM, SQL, C#, Unity [5]
Casting-temperature prediction from plant process data, with database integration and deployment tools. code

SoftmaxSGM | Python, PyTorch, Gymnasium [1]
Federated classification and safe RL on constrained CartPole, using custom TRPO updates. code RL notebook

EM Numerical Experiments | Python, Jupyter, Mathematica [2] [3]
Reproducible notebooks for cycloid trajectories and overspecified-EM dynamics. ICML code TMLR code

Interactive t-SNE and K-Means Visualizer | C++, Qt, OpenGL
Desktop application for inspecting embeddings and intermediate clustering iterations. code

SELECTED PUBLICATIONS & MANUSCRIPTS

Published

[1] **Z. Luo**, A. Upadhyay, S. B. Moon, and A. Hashemi. "First-Order Softmax Weighted Switching Gradient Method for Distributed Stochastic Minimax Optimization with Stochastic Constraints." **UAI 2026**. paper poster

[2] **Z. Luo** and A. Hashemi. "Characterizing Evolution in Expectation-Maximization Estimates for Overspecified Mixed Linear Regression." **TMLR 2026**. paper

[3] **Z. Luo** and A. Hashemi. "Unveiling the Cycloid Trajectory of EM Iterations in Mixed Linear Regression." **ICML 2024**. paper poster

[4] E. Cai, **Z. Luo**, S. Baireddy, J. Guo, C. Yang, and E. J. Delp. "High-Resolution UAV Image Generation for Sorghum Panicle Detection." **CVPR Workshops 2022**. paper video

[5] N. J. Walla, **Z. Luo**, B. Chen, Y. Krotov, and C. Q. Zhou. "Smart Ladle: AI-Based Tool for Optimizing Caster Temperature." **AISTech 2021**. paper poster

[6] **Z. Luo**, Y. Zhang, and L. Tan. "Multi-Level Random Sample Consensus Method for Improving Structured Light Vision Systems." **IEEE UEMCON 2020**. paper

[7] Y. Zhang, **Z. Luo**, J. Hou, L. Tan, and X. Guo. "Computer Vision Techniques for Improving Structured Light Vision Systems." **IEEE EIT 2020**. paper

Preprints & work in progress

[8] **Z. Luo**, A. Upadhyay, M. B. Sahin, S. B. Moon, A. Makur, and A. Hashemi. "Unified High-Probability Analysis of Stochastic Variance-Reduced Estimation." Preprint, 2026. NeurIPS reviewer ratings at decision: 5, 5, 4. preprint

[9] **Z. Luo**, M. B. Sahin, A. Upadhyay, B. Sharif, and A. Hashemi. "RAMPAGE: RAndomized Mid-Point for debiAsed Gradient Extrapolation." Preprint, 2026; submitted to AISTATS. preprint

[10] **Z. Luo** and A. Hashemi. "Connecting Score Matching, Maximum Likelihood, and Expectation-Maximization in Mixed Linear Regression." Preprint, 2026; submitted to TMLR. preprint

[11] **Z. Luo** and A. Hashemi. "Structural Properties, Cycloid Trajectories and Non-Asymptotic Guarantees of EM Algorithm for Mixed Linear Regression." Revised manuscript submitted after an *IEEE Transactions on Information Theory* revise-and-resubmit decision. preprint

[12] **Z. Luo**, D. Lee, A. Hashemi, and A. Makur. "Generalized Antithetic Variance Reduction: an Optimal Transport Approach." In preparation. slides

TEACHING & MENTORING

Teaching Assistant, Purdue ECE — **May 2022 – Aug. 2025**
Held office hours for ECE 69500: Optimization for Deep Learning slides course videos.
Led help sessions for ECE 20002 course videos; compiled an ECE 20001 practice-problem collection problems.

Graduate Student Mentor, Purdue Northwest ETIE. Mentored projects in sequential prediction and 3D reconstruction.

PROFESSIONAL SERVICE & LEADERSHIP

Peer reviewer. NeurIPS, ICLR, UAI, AISTATS, AAAI, IEEE ISIT, IEEE ICME; IEEE TSP, Stochastic Systems. ORCID

Leadership. Led a team that earned an ICM Honorable Mention certificate; coordinated outreach for 3,600+ students.

SELECTED AWARDS

NeurIPS Conference Travel Grant	**2025**
AIST Hunt-Kelly Outstanding Paper Award, third place award	**2022**
AIST Digitalization Applications Technology Best Paper Award award	**2021**